

Scientists Study How a Queen Mole-Rat's Scent Shapes Colony Life

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Naked mole-rats share work within their colonies. Some members search for food, while others care for the young. The queen has babies. A study found that a chemical scent produced by queens helps keep other colony members from fighting or reproducing.

Gary Lewin, a **neuroscientist** at the Max Delbrück Center in Berlin, Germany, studies how naked mole-rats use their senses. He previously discovered that they use scent to identify mole-rats from other colonies. That raised another question: do odors also matter within a colony?

Lewin and his team identified isopropyl myristate, a chemical produced only by queens among the mole-rats they studied. It triggered activity in the animals' **olfactory** cortex, the brain region that processes smells. This gave the researchers a clue that the animals could detect the chemical.

To investigate its effects, the team removed a queen from her colony. They then applied isopropyl myristate to the colony every day for three months. Lewin reported that there was no fighting or mating throughout that period. Mating is part of animal **reproduction**.

When the researchers stopped adding the odor, the mole-rats began fighting. A few weeks later, mating occurred and a new queen emerged. The experiment showed that the chemical could affect colony behavior even while the queen was absent. The team shared its findings in the journal *Nature*.

A separate experiment examined **reproduction** more closely. The researchers separated male and female pairs from the colony. They exposed some pairs to the queen's scent and left others without it. After four weeks, none of the females exposed to the scent had mated. In the group without the scent, five of six females were pregnant.

Blood tests revealed changes in female mole-rats' **hormones**, chemical signals that influence body functions. Isopropyl myristate lowered **progesterone** levels and raised **prolactin** levels. The researchers linked these changes to the prevention of pregnancy.

The team also found that queens produced the chemical only when pregnant or preparing to become pregnant. Together, the experiments help explain how a scent connects a queen's reproductive condition with the behavior and **reproduction** of other colony members.

Word Treasure

neuroscientist -- A scientist who studies the brain and nervous system.

olfactory -- Related to the sense of smell.

reproduction -- The process of having young.

hormones -- Chemical signals that influence body functions.

progesterone -- One of the hormones measured in the female mole-rats.

prolactin -- Another hormone measured in the female mole-rats.

Background you need

The olfactory cortex is a brain region that processes smells.

Hormones are chemical signals that influence body functions, including reproduction.

Break it down

Introduces colony roles and the study's main finding.

Explains Lewin's earlier research and the chemical's effect on the brain.

Describes a colony experiment with the queen absent.

Compares pairs exposed to the scent with pairs without it.

Connects hormone changes and the queen's reproductive condition to the findings.

Why it matters

The study helps explain how a chemical signal can connect a queen's reproductive condition with behavior and reproduction elsewhere in her colony.

Test what you remember

1. What earlier finding led Lewin to study odors within a colony?

- ☐ (A) Mole-rats use scent to identify outsiders.
- ☐ (B) Mole-rats use touch to choose a queen.
- ☐ (C) Mole-rats use sound to identify food.
- ☐ (D) Mole-rats use light to find their young.

2. What does the olfactory cortex do?

- ☐ (A) It processes smells.
- ☐ (B) It moves muscles.
- ☐ (C) It processes sounds.
- ☐ (D) It digests food.

3. What did researchers do after removing the queen?

- ☐ (A) They returned the queen every night.
- ☐ (B) They tested a new odor every week.
- ☐ (C) They added the chemical every day.
- ☐ (D) They moved the colony every month.

4. What did Lewin report during the three-month chemical treatment?

- ☐ (A) Frequent fighting but no mating.
- ☐ (B) No fighting but frequent mating.
- ☐ (C) Frequent fighting and frequent mating.
- ☐ (D) No fighting and no mating.

5. What was observed after four weeks in the pairs exposed to the scent?

- ☐ (A) Every female had become a queen.
- ☐ (B) All pairs had rejoined the colony.
- ☐ (C) None of the females had mated.
- ☐ (D) Five of six females were pregnant.

6. How did the chemical affect the two hormones measured?

- ☐ (A) Both progesterone and prolactin fell.
- ☐ (B) Progesterone rose and prolactin fell.
- ☐ (C) Both progesterone and prolactin rose.
- ☐ (D) Progesterone fell and prolactin rose.

Answer key (try the quiz first!): 1: A 2: A 3: C 4: D 5: C 6: D

1. Gary Lewin: He reported that the colony showed no fighting or mating during the three months when researchers added the queen's scent.